Minchan Kim

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RESEARCH INTERESTS

Learning from relations and feedback, across three settings: **Graph Learning** · **Recommender Systems** · **AI Agents**. Specific interests include Graph Representation Learning, Relational Deep Learning, Knowledge Graph Reasoning, and LLM Reasoning & Interpretability.

EDUCATION

Seoul National University — Mar. 2024 – Present
Ph.D. in Data Science — Seoul, Republic of Korea

- Advisor: Prof. Hyunwoo Park (DIAL Lab)

Seoul National University of Science & Technology — Sept. 2022 – Feb. 2024
M.S. in Data Science — Seoul, Republic of Korea

- GPA: 4.45 / 4.5 | Advisor: Prof. Hakyeon Lee (D&BI Lab)

Seoul National University of Science & Technology — Mar. 2020 – Aug. 2022
B.S. in Computer Science and Engineering — Seoul, Republic of Korea

Seoul National University of Science & Technology — Mar. 2017 – Aug. 2022
B.B.A. in Business Administration — Seoul, Republic of Korea

- GPA: 4.25 / 4.5 | Rank: 1 / 30

PUBLICATIONS

[*] *These authors contributed equally.*

Working Papers

1. **Minchan Kim**[*], Sungho Bae[*], Youngjin Kim, and Hyunwoo Park. "Where to Start, How to Move: Analyzing Curricula under GRPO for Long-Horizon LLM Agents." (Under review.) arXiv:2610.19349.
2. **Minchan Kim**[*], Joonsung Kwon[*], Jeehong Kim, and Hyunwoo Park. "Relational Injection Attack Against Relational Deep Learning." (Under review.)

Journal Articles

1. **Minchan Kim**[*], Yanjie Zhao[*], Jaeseong Ju, Jaeeun Seo, and Hyunwoo Park. "InteractiveKG: Transparent and User-Controllable Knowledge Graph Reasoning." *Information Visualization*, 2026. (Special Issue from PacificVis 2026 Vis Meets AI Workshop.)
2. Jeehong Kim[*], **Minchan Kim**[*], Youngseok Hwang, Sungho Bae, Deuk Jae Cho, Wonhee Lee, and Hyunwoo Park. "STGVAD: Spatio-Temporal Graph-Based Vessel Behavior Anomaly Detection." *IEEE Access*, 2025.
3. **Minchan Kim** and Hakyeon Lee. "Treasure Hunting in the Talent Ocean: Automating Talent Acquisition for Competent Developers from GitHub." *Annals of Data Science*, 2026.

Conference Papers

1. **Minchan Kim**, Jungmin Hwang, and Hyunwoo Park. "What Gets Measured Gets Managed: Sign-aware Recommendation Needs Sign-aware Evaluation." *NeurIPS 2026 Evaluations and Datasets Track*, Sydney, Australia, 2026.

Workshop Papers

1. Jeehong Kim[*], Youngseok Hwang[*], **Minchan Kim**, Sungho Bae, and Hyunwoo Park. "Spatio-Temporal Graphs Beyond Grids: Benchmark for Maritime Anomaly Detection." *NeurIPS 2025 Workshop on AI for Science*, San Diego, CA, USA, 2025.

2. Jeehong Kim*, **Minchan Kim***, Jaeseong Ju, Youngseok Hwang, Wonhee Lee, and Hyunwoo Park. "Adaptive Sparsified Graph Learning Framework for Vessel Behavior Anomalies." *AAAI 2025 Workshop and Challenge on Anomaly Detection in Scientific Domains*, Philadelphia, PA, USA, 2025.

PATENTS

1. "Developer Head-Hunting Candidate Automation System." (Domestic) Application No. 0060095, 2023.

RESEARCH PROJECTS

Fraud Detection System (FDS) Modeling Optimization for Overseas Remittance Using AI Jul. 2025 – Present
MOIN

Vessel Behavior Anomaly Detection Oct. 2024 – Dec. 2025
Korea Research Institute of Ships & Ocean Engineering (KRISO)

- Resulting publications: STGVAD (*IEEE Access*, 2025), with workshop papers at NeurIPS 2025 and AAAI 2025.

Center for Optimizing Hyperscale AI Models and Platforms (ERC) Mar. 2024 – Present
National Research Foundation of Korea (NRF) – Engineering Research Center

- Resulting publication: InteractiveKG (*Information Visualization*, 2026).

Personalized Knowledge State Analysis & Question Recommendation with Knowledge Tracing Jan. 2024 – Feb. 2024
Onuii

ML-based Information Processing Analysis for B2B Business Environment Analysis Sept. 2023 – Feb. 2024
LG Electronics

ML-based Graph Learning & Link Prediction for Text-based Technology Trend Analysis Jul. 2022 – May 2023
Hyundai Motor Company

Nowcasting of Artificial Intelligence with Open Source Project Data Mar. 2022 – Feb. 2024
Ministry of Science and ICT

- Resulting publication: "Treasure Hunting in the Talent Ocean" (*Annals of Data Science*, 2026).

University Industry-Academia Research Capability Big Data Analysis Sept. 2021 – Dec. 2021
Seoul National University of Science & Technology

AWARDS & GRANTS

SNU Medical AI Scholarship 2025
Seoul National University

BK21 Scholarship 2022 – 2023, 2025
National Research Foundation of Korea (NRF)

Electronics Manufacturing Specialist Training Program 2022 – 2023
Korea Institute for Advancement of Technology (KIAT)

Grand Prize, Idea Hackathon for Customs Clearance Reading Using AI 2022
National IT Industry Promotion Agency (NIPA)

OTHER EXPERIENCE

Supporting Victims of Cyberbullying & Reducing Social Media Violence (TLAB) 2022 – Present

- Collected data via Twitter and YouTube APIs; built intuitive visualizations with matplotlib, seaborn, and pyvis.

Improving Medicine Access for the Visually Impaired 2022

- Implemented OCR with YOLOv5 and clovaai text-recognition models to extract text from medicine labels.

VOLUNTEER

Rescue Organization for Stray Cats in Redevelopment Areas 2021 – Present

Overseas Volunteer, Vietnam (Seoul National University of Science & Technology) 2020

Entrepreneurial Education Corps for Middle & High School Students 2020

REFERENCES

Prof. Hyunwoo Park Ph.D. Advisor
Associate Professor, Graduate School of Data Science, Seoul National University hyunwoopark@snu.ac.kr

Prof. Hakyeon Lee M.S. Advisor
Professor, Industrial Engineering & Data Science, Seoul National University of Science & Technology hylee@seoultech.ac.kr